# BMS_MASTER

## Variant: DRAFT

# 2026-10-21

# Rev + (Unreleased)


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## TOP VIEW



## BOTTOM VIEW



## NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

## DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

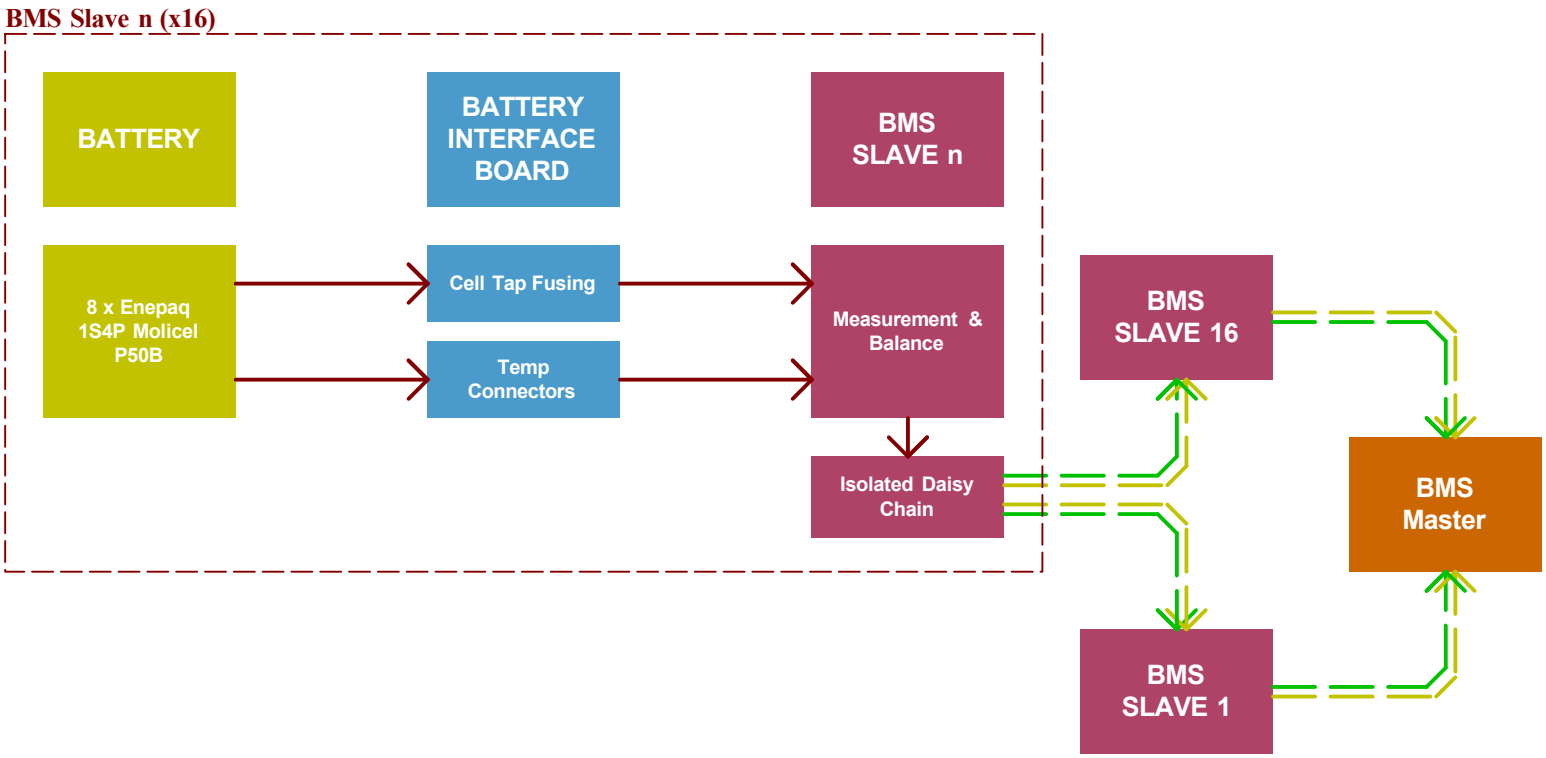
**DESIGN NOTE: Example text for critical design notes.**

**LAYOUT NOTE: Example text for critical layout guidelines.**

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| | | | | |
|---|---|---|---|---|
| Comments: | Company: Blue Devil Motor Sports | | Variant: DRAFT | Git Hash: c3341ef |
| | Board Name: BMS_MASTER | | Project Name: BMS Master | |
| Sheet Title: | File Name: bms_master.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: / | | Reviewer: | Size: A3 | Sheet: 1 of 8 |

# [2] Block Diagram



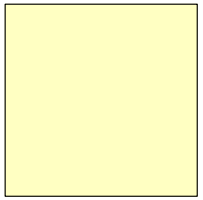

**Target specifications:**

| | |
|---|---|
| **Input voltage:** | ?? - ?? V |
| **Spec 2** | ?? |
| **Spec 3** | ?? |
| **Spec 4** | ?? |

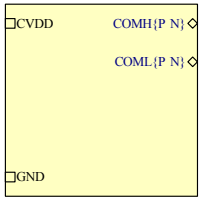
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|---|---|---|---|---|
| Comments: | Company: Blue Devil Motor Sports | | Variant: DRAFT | Git Hash: c3341ef |
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| Sheet Title: Block Diagram | File Name: Block Diagram.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Block Diagram/ | | Reviewer: | Size: A4 | Sheet: 2 of 8 |

# [3] Project Architecture

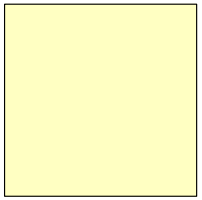
**Com. Chip**



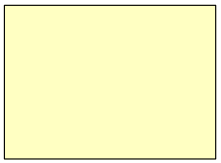
**Daisy Chain**




**Connectors**



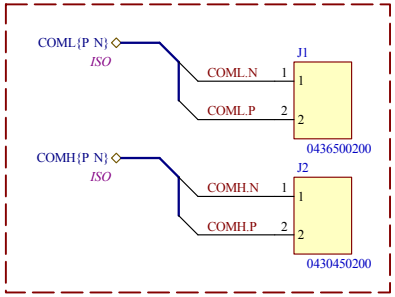
**Mechanical**



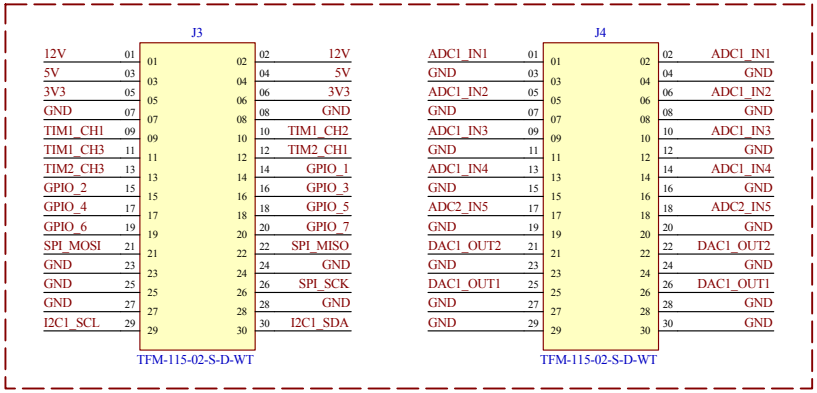
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|---|---|---|---|---|
| Comments: | Company: Blue Devil Motor Sports | | Variant: DRAFT | Git Hash: c3341ef |
| | Board Name: **BMS_MASTER** | | Project Name: **BMS Master** | |
| Sheet Title: Project Architecture | File Name: Project Architecture.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/ | | Reviewer: | Size: **A4** | Sheet: **3** of **8** |

# [4] Connectors

## BMS Comminication Connectors

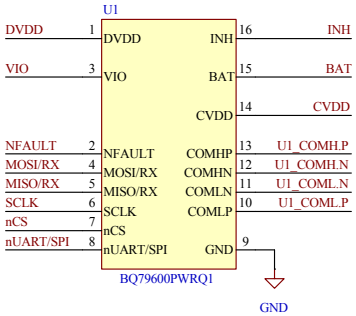


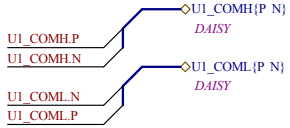

## Board to Board Connector




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| | Board Name: **BMS_MASTER** | | Project Name: **BMS Master** | |
| Sheet Title: Connectors | File Name: connectors.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Connectors/ | | Reviewer: | Size: **A4** | Sheet: **4** of **8** |

# [5] Com. Chip

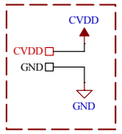


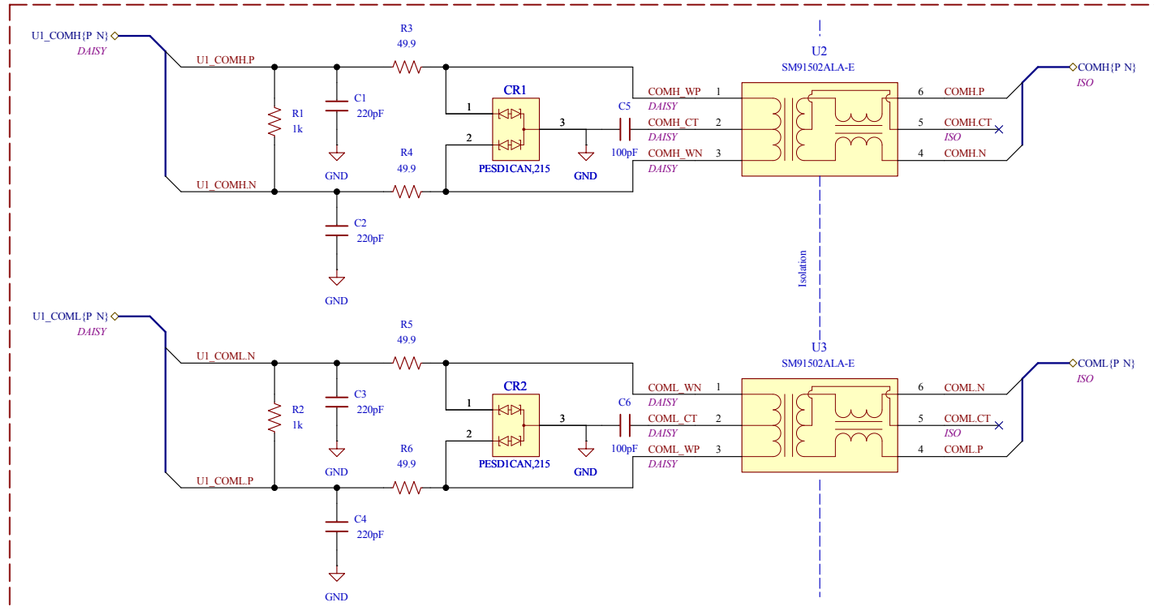

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|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: DRAFT | Git Hash: c3341ef |
| | Board Name: **BMS_MASTER** | | Project Name: **BMS Master** | |
| Sheet Title: Com. Chip | File Name: com_chip.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Com. Chip/ | | Reviewer: | Size: **A4** | Sheet: **5** of **8** |

# [6] Daisy Chain

**Power**




**Communication Isolation**




DESIGN NOTE: DAISY CHAIN ISOLATION

1. Transformer isolation per SLUSE81F Table 8-5 / Fig 8-9: R_TERM 1 kΩ, R_SERIES 49.9 Ω, C_BYPASS 220 pF/50 V C0G, centre-tap cap 100 pF to GND, TVS PESD1CAN on chip side.
2. Cable-side centre taps (U2-5, U3-5) are left NC per Fig 8-9.
3. Stack device: RX tied to CVDD, TX and NFAULT left NC (SLUSE81F 8.2.1.2.6).
4. U3 is drawn with chip-side pins 1/3 opposite to U2. Polarity is preserved end to end (P->P, N->N).
5. Ring architecture: COMH (J2) goes up the stack, COML (J1) goes down. J1 = 43650-0200 single-row, J2 = 43045-0200 dual-row so a cable cannot connect COMH to COMH.
6. Use the same transformer PN on both ends of every link, including the master (TI SLVAEP4 5.1).

**LAYOUT NOTE: DAISY CHAIN (TI SLUSE81F 10.1.4, Fig 10-3)**

**1. COMH and COML pairs: short, straight, same layer, parallel, length matched. No vias in the pairs.**
**2. R_TERM and C_BYPASS close to U1 pins 40-43.**
**3. Solid GND under the chip-side parts (R, C, TVS) is required for the bypass and TVS returns. Keep unrelated traces out of this area, and never route a COM pair across a plane seam.**
**4. No copper on ANY layer under the transformer bodies or between the transformer cable side and J1/J2.**
**5. Transformer barrier: 7.11 mm pad gap meets EV.7.5.7 300-600 V coated creepage (4.0 mm) and clearance (3.0 mm). Keep >= 4.0 mm on all layers. No plated holes, vias or pours inside the isolation zone.**
**6. Place each transformer next to its connector.**
**7. TP11-TP14: bare SMD pads on the trace, no stubs.**

CAUTION: ISOLATION SPACING (FSAE 2027 EV.8.3.2, EV.7.5.7)

1. EV.8.3.2: the BMS must have galvanic isolation at every Module-to-Module boundary. U2/U3 provide it. All slaves are identical, so every barrier is designed for the worst case: full pack voltage, <128S x 4.2 V = 537.6 V> max (EV.3.3.2 limit: 600 V DC).
2. EV.7.5.7 spacing, 300-600 V DC row:
Clearance (through air): 3.0 mm
Creepage, general (uncoated): 20.0 mm
Creepage, under conformal coating: 4.0 mm
3. This board IS conformally coated, so the 4.0 mm creepage and 3.0 mm clearance apply. Uncoated it would NOT comply (7.11 mm < 20 mm).
4. Rule applies on the same layer AND between layers. Keep >= 4.0 mm on every layer between chip-side copper (inner GND pours, vias, TVS / CT-cap pads) and cable-side copper (U2/U3 pins 4-6, J1/J2). Actual: 7.11 mm transformer pad gap (SM91502ALA land pattern).
5. No plated holes, vias or copper pours inside the isolation zone.
6. Mark the zone on top AND bottom silkscreen (boundary lines + "ISOLATION ZONE"), per EV.7.5.6 / EV.7.5.7a.
7. U2/U3 isolation rating and the daisy-chain cable insulation must be rated >= max TS voltage (EV.5.2.1, EV.7.3.2).
8. Keep a spare coated board or photos for inspection (EV.7.5.8).

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| Sheet Title: Daisy Chain | File Name: daisy_chain.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Daisy Chain/ | | Reviewer: | Size: **A3** | Sheet: **6** of **8** |

# [8] Mechanical

| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: DRAFT | Git Hash: c3341ef |
| | Board Name: **BMS_MASTER** | | Project Name: **BMS Master** | |
| Sheet Title: Mechanical | File Name: mechanical.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Mechanical/ | | Reviewer: | Size: **A4** | Sheet: **8** of **8** |

# [7] Revision History

| | | | | | |
|---|---|---|---|---|---|
| Comments: | Company: **Blue Devil Motor Sports** | | Variant: DRAFT | Git Hash: c3341ef | |
| | Board Name: **BMS_MASTER** | | Project Name: **BMS Master** | | |
| Sheet Title: Revision History | File Name: Revision History.kicad_sch | Designer: Aidan Brzezinski | Date: 2026-10-20 | Revision: + (Unreleased) | |
| Sheet Path: /Revision History/ | | Reviewer: | Size: **A4** | Sheet: **7** of **8** | |